DCIT 316 — RESPONSIBLE SOCIAL MEDIA MINING FOR PUBLIC GOOD | FINAL RESEARCH DEFENCE

Can We Trust the Confidence?

Per-Class Calibration of Twi and Hausa Hate-Speech Classifiers Under Class Imbalance

TARGET DATASET

AfriHate Benchmark
(Muhammad et al., NAACL 2025)
Twi ('twi') & Hausa ('hau')

MODEL ARCHITECTURE

Word + Char TF-IDF
Balanced Logistic Regression
Deterministic Seed: SEED=42

CORE CONTRIBUTION

Per-Class ECE & Blinding Rates
Temperature Scaling Inversion
Triage Risk-Coverage Frontier

REPOSITORY & CODE

GitHub: NEMESIS-7/CMSMM_finalproject
Primary Notebook: calibration_study.ipynb
Includes Data & Model Cards

Student Submission & Oral Defence | Course: DCIT 316 — Academic Year 2025/2026

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PROBLEM DEFINITION & OPERATIONAL CONTEXT

Why Confidence Calibration Matters in Content Moderation

The Triage Pipeline Promise

- **High-Volume Platforms:** Human moderators cannot review millions of daily posts. Automated triage divides the workload based on predicted confidence.
- **Automated Tier (p >= tau):** High-confidence predictions are automatically actioned (removed, flagged, or cleared) without human intervention.
- **Human Escalation (p < tau):** Low-confidence predictions are routed to human review queues for careful scrutiny.
- **The Core Premise:** This entire system assumes confidence reflects true correctness probability. If a model claims 90% confidence, it must be right ~90% of the time.

The Failure Mode: 'Moderator Blinding'

- **The Danger of Confident Errors:** When a model is confidently wrong, it files mistakes under 'Trust Me' and routes them away from human oversight.
- **Active Misdirection:** The system does not merely fail—it blinds the human reviewer by suppressing the very posts that urgently require correction.
- **Directional Harms in African Contexts:** Misclassifying Hate as Normal allows harmful abuse to propagate; misclassifying Normal as Hate/Abuse censors innocent Ghanaian/Nigerian speakers.
- **The Calibration Blindspot:** Standard evaluations report only overall accuracy or aggregate loss, completely hiding class-specific confidence distortions.

CRITICAL DISTINCTION: ACCURACY VS. CALIBRATION

- Accuracy measures overall correctness ('How often is the model right?').
- Calibration measures probability reliability ('When the model claims 80% certainty, is it actually right 80% of the time?').
- A classifier can achieve high accuracy while remaining dangerously overconfident on its critical errors.

RESEARCH QUESTIONS & HYPOTHESES

Investigating Per-Class Calibration Under Inverted Skew

RQ1: Trust Across Classes

Is a hate-speech classifier's confidence equally trustworthy across classes, or does calibration degrade substantially on the minority class?

Theoretical Framing:

Hypothesis: Confidence is not uniform; minority classes suffer degraded calibration due to sparse training representations.

RQ2: Mitigation Efficacy

Does standard post-hoc temperature scaling recover calibration uniformly across all classes, or does it fix aggregate loss while leaving specific classes worse off?

Theoretical Framing:

Hypothesis: Scalar temperature scaling optimizes for the majority class, risking per-class calibration inversion on the minority.

RQ3: Cross-Language Inversion

Does the worst-calibrated class follow the rare class when comparing Twi (rare: Normal) against Hausa (rare: Hate)?

Theoretical Framing:

Hypothesis: The miscalibration failure mode flips between languages, tracking whichever class is underrepresented.

A NATURAL EXPERIMENT: INVERTED CLASS PROFILES ACROSS TWO AFRICAN LANGUAGES

- Identical Task & Schema: 3-class classification ('Abuse', 'Hate', 'Normal') using identical preprocessing and features.
- Twi (Ghana): Abuse is dominant (70.5%), while Normal (15.2%) and Hate (14.3%) are rare minority classes.
- Hausa (Nigeria/West Africa): Normal is dominant (56.7%), while Abuse (33.5%) is mid-frequency and Hate (9.8%) is rare.
- Research Value: If miscalibration tracks rarity, the failure mode will cleanly flip between Twi and Hausa.

THEORETICAL FOUNDATIONS & LITERATURE MATRIX

Synthesizing Calibration Theory and African NLP Benchmarks

Guo et al. (ICML 2017)

On Calibration of Modern Neural Networks

Key Insights:

• Demonstrated modern networks are systematically overconfident.
• Formalized Expected Calibration Error (ECE) and Reliability Diagrams.
• Introduced scalar Temperature Scaling as a simple post-hoc fix.

Relevance: Directly provides our evaluation metrics and mitigation baseline.

Ahuja et al. (EMNLP 2022)

Calibration of Massively Multilingual Models

Key Insights:

• Multilingual calibration degrades on low-resource and typologically distant languages.
• Acceptable accuracy masks severe confidence errors.
• Post-hoc mitigations fail to close the language gap.

Relevance: Motivates evaluating African languages empirically rather than assuming transfer.

Islam et al. (2021)

Calibration for Long-Tailed Recognition

Key Insights:

• Scalar temperature scaling is suboptimal under class imbalance.
• Majority classes dominate validation loss, skewing scalar T.
• Minority classes carry disproportionate calibration error.

Relevance: Provides direct theoretical motivation for per-class ECE measurement.

Muhammad et al. (NAACL 2025)

AfriHate: African Hate Speech Benchmark

Key Insights:

• First extensive hate speech collection across 15 African languages.
• Native-speaker annotations from Ghanaian and Nigerian contexts.
• Documents extreme skew and bidirectional moderation failures.

Relevance: Supplies our benchmark data, ground-truth labels, and baseline framing.

AfriHate: Verified Class Counts and Natural Experiment Design

Step 0 Pre-Verification Checks (Passed)

- **1. True Label Strings:** Verified via df['label'].unique() -> ['Abuse', 'Hate', 'Normal']. Avoided hardcoded prose assumptions.
- **2. Test Cell Viability:** Rarest test cell check confirmed N >= 30 (Twi Hate: 100, Normal: 106). Both languages fully viable.
- **3. Schema Parity:** Aligned common columns ['id', 'tweet', 'label'] across both subsets.

Subset	Train	Validation	Test	Dominant Class
Twi	2,564	639	698	**Abuse (70.5%)**
Hausa	4,566	1,029	1,049	**Normal (56.7%)**

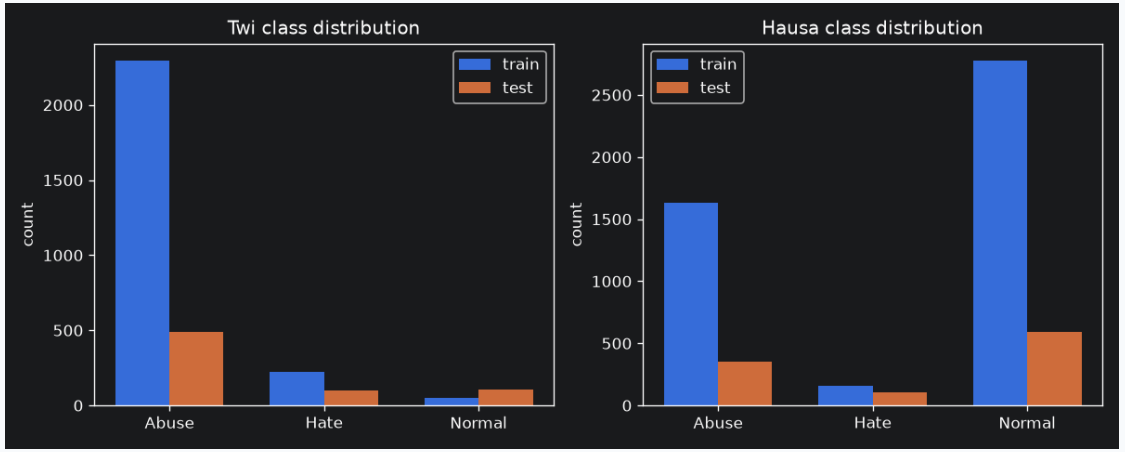


Figure 1: Verified class distributions across Twi and Hausa splits. Notice the clear natural inversion: Twi is overwhelmingly dominated by Abuse (70.5%), whereas Hausa is dominated by Normal speech (56.7%).

THE DECEPTIVE ACCURACY TRAP

- On Twi, a dummy majority classifier predicting 'Abuse' on every post achieves 70.5% accuracy while detecting exactly 0% of Normal or Hate speech!
- Reporting accuracy as a headline metric is mathematically deceptive.
- Macro-F1 is our sole headline performance metric.

METHODOLOGY & CALIBRATION ARCHITECTURE

Feature Pipeline, Non-Parametric Binning, and Mitigation Setup

1. Feature Engineering & Model

Tailored to African Morphology:

• Word n-grams (1-2): Captures semantic unigrams and phrases with sublinear TF scaling.
• Character n-grams (2-4): Word-boundary char n-grams capture agglutinative affixes and social code-switching.
• Balanced Logistic Regression: Inverse-frequency class weights counteract majority-class gradient drowning.
• Deterministic Execution: Fixed SEED=42 ensures exact scientific reproducibility across all runs.

2. Adaptive ECE with Bootstrap CIs

One-vs-Rest Equal-Mass Binning:

• Equal-Mass (Adaptive) Quantiles: Standard equal-width bins produce empty cells on skewed classes; equal-mass ensures every bin has equal sample mass (M=10 bins).
• Formal Formulation:
ECE_c = SUM (|B_m| / N) * |p_bar_m - y_bar_m|
• 95% Bootstrap CIs: 1,000 resamplings quantify uncertainty arising from small minority cells (N=100-106).
• Reliability Curves: Diagonal tracks perfect calibration; dot sizes encode bin support.

3. Blinding Quantity & Scaling

Operationalizing Triage Risk:

• The Blinding Quantity (tau >= 0.70):
Blinding_c = (Class c Errors with max p >= 0.70) / (Total Class c Errors)
• Operational Consequence: Measures the fraction of errors filed under 'Trust Me' that bypass human review.
• Post-Hoc Temperature Scaling:
p_hat = softmax(z / T)
Scalar T fit on validation NLL via bounded L-BFGS-B (bounds [0.01, 10.0]). Test set strictly held out.

EMPIRICAL FINDINGS: CLASSIFICATION PERFORMANCE

Test Set Evaluation Reveals Severe Minority Class Recall Collapse in Twi

Language	Class	Precision	Recall	F1	Support
Twi	Abuse	0.87	0.98	0.92	492
Twi	Hate	0.56	0.70	0.62	100
Twi	Normal	0.75	**0.14**	**0.24**	106
Twi	**Macro Avg**	**0.73**	**0.61**	**0.59**	**698**
Hausa	Abuse	0.75	0.72	0.74	351
Hausa	Hate	0.82	0.53	0.65	103
Hausa	Normal	0.80	0.87	0.83	595
Hausa	**Macro Avg**	**0.79**	**0.71**	**0.74**	**1,049**

Key Classification Insights:

- **Twi Macro-F1 (0.59) vs Hausa (0.74):** Hausa model shows significantly stronger and more balanced discrimination across all three categories.
- **The Twi Normal Collapse (Recall = 0.14):** The Twi model detects only 14% of innocent Normal posts. 86% of innocent speech is misclassified as Abuse or Hate.
- **Majority-Class Bias:** Twi achieves 98% recall on Abuse by aggressively defaulting to it, creating a severe over-censorship hazard for Ghanaian users.

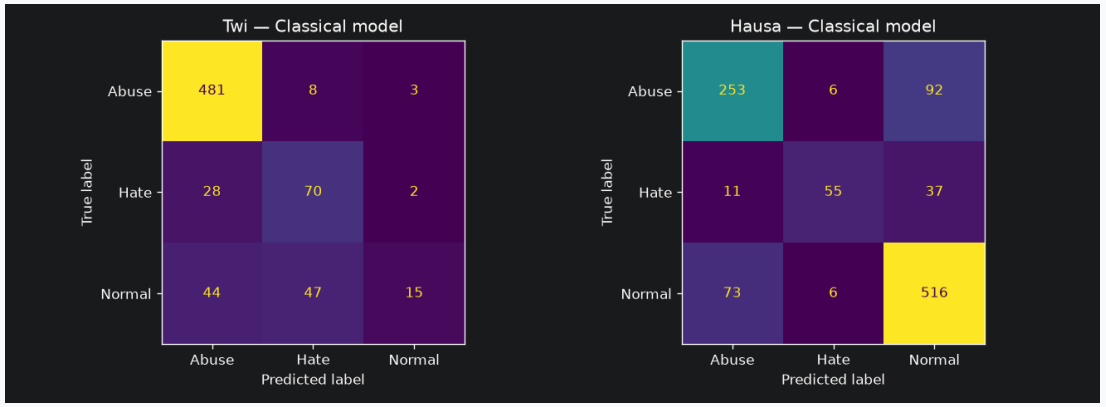


Figure 2: Normalized confusion matrices. In Twi, Normal posts are heavily misrouted to Abuse (0.42) and Hate (0.43). In Hausa, errors are symmetrically distributed between Abuse and Normal.

EMPIRICAL FINDINGS: PER-CLASS CALIBRATION (ECE)

Twi Calibration is 2x Worse; Dominant Class Exhibits Highest Calibration Error

Language	Class	ECE (Equal-Mass)	95% Bootstrap CI
Twi	**Abuse (Dominant)**	**0.0848**	**[0.065, 0.106]**
Twi	Hate	0.0828	[0.065, 0.103]
Twi	Normal (Rare)	0.0643	[0.050, 0.091]
Twi	**Macro Average**	**0.0773**	**—**
Hausa	Abuse	0.0415	[0.034, 0.066]
Hausa	Hate (Rare)	0.0243	[0.018, 0.039]
Hausa	**Normal (Dominant)**	**0.0468**	**[0.040, 0.075]**
Hausa	**Macro Average**	**0.0375**	**—**

Surprising Theoretical Findings:

- **Twi Calibration is 2x Worse:** Macro ECE for Twi (0.077) is double that of Hausa (0.038), confirming Ahuja et al.'s finding on low-resource degradation.
- **The Dominant Class Paradox:** Contrary to our initial hypothesis, the dominant class carries the HIGHEST ECE in both languages (Abuse in Twi: 0.085; Normal in Hausa: 0.047).
- **Why Dominant Classes Miscalibrate:** Abundant training data causes the model to assign inflated confidence. Rare class predictions, when made, are appropriately hedged.

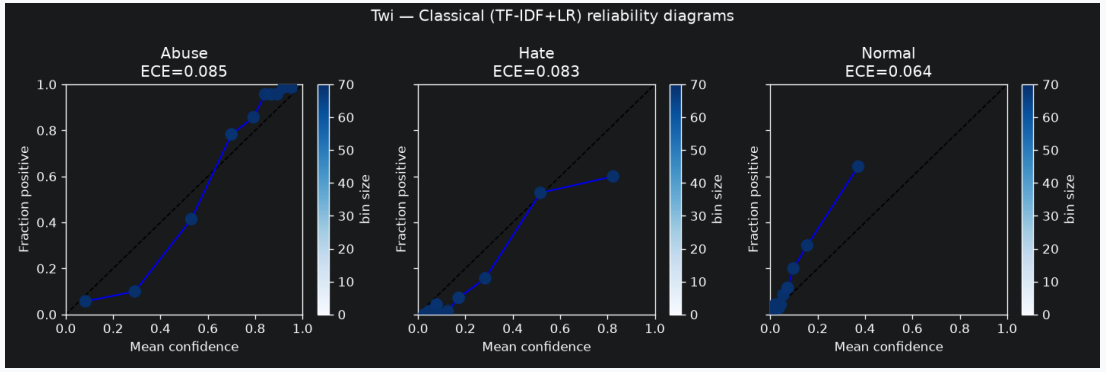


Figure 3: Twi reliability diagrams before (blue) and after (orange) temperature scaling across Abuse, Hate, and Normal classes. Notice the divergence in the Normal class.

CENTRAL FINDING: THE CALIBRATION TRAP

Temperature Scaling Improves Aggregate ECE but Worsens Twi Normal by +42%

Language	Optimal T	Class	ECE Before	ECE After	Delta / Outcome
Twi	0.7106	Abuse	0.0848	0.0426	-0.042 (Improved)
Twi	0.7106	Hate	0.0828	0.0604	-0.022 (Improved)
Twi	**0.7106**	**Normal**	**0.0643**	**0.0910**	**+0.027 (+42% WORSENED!)**
Twi	**—**	**Macro Avg**	**0.0773**	**0.0647**	**-0.013 (Deceptive Gain)**
Hausa	0.7907	Abuse	0.0415	0.0361	-0.005 (Improved)
Hausa	0.7907	Hate	0.0243	0.0109	-0.013 (Improved)
Hausa	0.7907	Normal	0.0468	0.0219	-0.025 (Improved)
Hausa	**—**	**Macro Avg**	**0.0375**	**0.0230**	**-0.015 (Uniform Benefit)**

Cross-Language Contrast:

- Hausa's balanced distribution allows temperature scaling to work as intended: all classes improve uniformly (Macro ECE drops from 0.038 to 0.023).
- This proves that post-hoc scaling failure is not a flaw of the language, but a direct statistical pathology of severe class imbalance interacting with scalar optimization.
- Core Takeaway: Never deploy post-hoc calibration without auditing per-class ECE.

THE MECHANISM OF THE TRAP

- **Why T < 1.0?** Optimal temperatures are 0.71 (Twi) and 0.79 (Hausa). T < 1 divides logits by a fraction, sharpening output probabilities toward extremes.
- **Validation Loss Domination:** In Twi, validation NLL is 70% Abuse instances. The optimizer fits T to satisfy majority Abuse loss.
- **The Minority Disaster:** Sharpening logits concentrates probability mass on Abuse/Hate. When the model encounters a Normal post, it now misclassifies it with EVEN HIGHER confidence than before!
- **The Illusion of Safety:** An ML engineer looking only at overall ECE (0.077 -> 0.065) would conclude calibration succeeded. In reality, over-censorship confidence increased by 42%!

OPERATIONAL RISK & ERROR ANALYSIS

Four in Ten Errors are Confidently Routed Away from Human Scrutiny

Moderator Blinding Rates (tau >= 0.70)

Fraction of class errors made with maximum confidence >= 0.70:

Language	Class	Total Errors	Blinding Rate
Twi	Abuse	11	**45.5% (5/11)**
Twi	Hate	30	**33.3% (10/30)**
Twi	**Normal**	**91**	**39.6% (36/91)**
Hausa	Abuse	98	**26.5% (26/98)**
Hausa	Hate	48	**31.2% (15/48)**
Hausa	Normal	79	**32.9% (26/79)**

OPERATIONAL HAZARD: TWI NORMAL OVER-CENSORSHIP

- Out of 91 total Normal errors in Twi, 36 are made at >= 0.70 confidence (39.6% blinding rate).
- In an automated triage pipeline, 36 harmless Ghanaian posts would be automatically censored without human scrutiny.
- The system quietly files innocent speech under 'Trust Me', actively deflecting review away from over-censorship mistakes.

Confident Error Patterns (tau >= 0.70)

Twi Confident Errors (51 total / 7.3% of test set):
- Normal -> Hate: 18 posts (Over-censorship of harmless speech)
- Normal -> Abuse: 18 posts (Over-censorship of harmless speech)
- Hate -> Abuse: 10 posts (Between-toxic misclassification)
- Abuse -> Hate: 3 posts | Abuse -> Normal: 2 posts

Striking Symmetry in Twi:
Exactly 18 Normal posts are confidently mislabeled as Hate and 18 as Abuse. The classifier lacks a stable internal representation of Normal Twi text; it arbitrarily projects innocent posts into toxic feature spaces with high confidence.

Hausa Confident Errors (67 total / 6.4% of test set):
- Abuse -> Normal: 25 posts (Under-detection)
- Normal -> Abuse: 25 posts (Over-censorship)
- Hate -> Normal: 11 posts (Critical safety miss)
- Symmetry: Exactly 25 posts each way along the Abuse-Normal boundary.

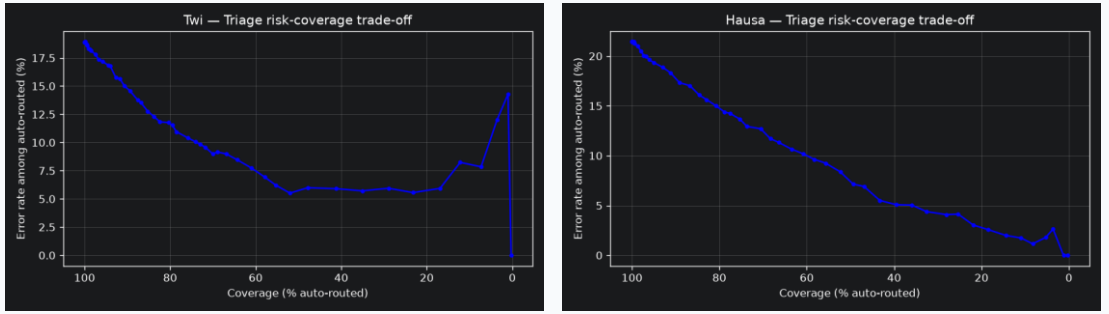
DEPLOYMENT ANALYSIS: RISK-COVERAGE TRADE-OFFS

Workload Reduction vs. Hidden Errors Across Confidence Thresholds

Language	Threshold (tau)	Auto-Routed	Coverage	Precision	Hidden Errors
Twi	0.50	656 / 698	**94.0%**	**83.2%**	**110**
Twi	0.70	514 / 698	**73.6%**	**90.1%**	**51**
Twi	0.90	182 / 698	**26.1%**	**94.5%**	**10**
Hausa	0.50	971 / 1,049	**92.6%**	**81.4%**	**181**
Hausa	0.70	652 / 1,049	**62.2%**	**89.7%**	**67**
Hausa	0.90	164 / 1,049	**15.6%**	**98.2%**	**3**

Operational Trade-off Dilemmas:

- **Aggressive Automation (tau = 0.50):** Reduces human workload by 93-94%, but permits 110 hidden errors in Twi and 181 in Hausa to slip past review.
- **Moderate Tier (tau = 0.70):** Auto-routes ~62-74% of posts with ~90% precision. However, 51 Twi errors remain hidden—36 of which are innocent speech.
- **Conservative Tier (tau = 0.90):** Hausa achieves 98.2% precision with only 3 hidden errors. But Twi still hides 10 errors while routing only 26% of posts automatically.



Figures 4A & 4B: Risk-coverage frontiers for Twi (left) and Hausa (right). Hausa's high-confidence tier is significantly cleaner, while Twi suffers persistent error leakage.

ETHICS, DATA GOVERNANCE & RESPONSIBLE AI

Safe Data Handling, Diagnostic Scope, and Transparent AI Disclosures

1. Data Privacy & Zero Redistribution

Conservative Handling Under Apache 2.0:

- Strict Privacy: No raw tweet text is committed to the GitHub repo, printed in notebooks, or stored in artifacts.
- User Anonymity: No user handle tracing, re-identification, or account scraping was performed.
- Derived Outputs Only: Only aggregate confusion matrices, ECE values, and reliability plots are published.

2. Diagnostic Instrument Designation

Strict Operational Boundary:

- Research Tool Only: These models are diagnostic instruments built to study calibration dynamics under skew.
- NOT Deployable: These classifiers must NEVER be deployed in live moderation pipelines.
- Harm Prevention: Live deployment in Ghana would result in severe over-censorship of harmless Twi speech.

3. Sampling & Base Rate Limitations

AfriHate Benchmark Context:

- Keyword & Handle Sampling: The dataset was compiled via targeted queries, not random sampling.
- Not Real-World Base Rates: Class proportions (70% Abuse in Twi) reflect collection filters, not the true prevalence of toxic speech in West Africa.
- Contextual Caveat: Findings evaluate classifier behavior on this distribution, not societal base rates.

4. Transparent AI Disclosure

Methodological Verification:

- Assistance Used: Claude / Gemini used for code scaffolding, plotting utilities, and report drafting.
- Independent Verification: 100% of numerical results, bootstrap CIs, and figures independently verified against deterministic code runs (SEED=42).
- Integrity: No AI was used to fabricate data, citations, or labels.

Critical Analysis of Theoretical Hypotheses and Scope Reductions

1. Single Model Family (Scope Reduction)

Documented Compute Constraint:

- The original proposal envisioned contrasting classical models against a fine-tuned multilingual transformer (AfroXLMR).
- Due to local CPU-only compute constraints and extreme training latency, the transformer was omitted as a documented scope cut.
- Impact: The accuracy-vs-calibration contrast across model families remains an open question for future GPU work.

2. Sampling Uncertainty in Small Cells

Bootstrap Confidence Interval Width:

- With N=100 (Hate) and N=106 (Normal) in Twi test, bootstrap CIs are wide (Normal CI: [0.050, 0.091]).
- Overlapping CIs mean fine-grained rank ordering between Twi Hate and Normal must be interpreted cautiously.
- Larger test splits are necessary to tighten statistical bounds.

3. Uniform vs Class-Conditional Scaling

Single Scalar Limitation:

- We evaluated standard single-scalar temperature scaling (Guo et al., 2017).
- The failure on Twi Normal demonstrates that a single scalar cannot reconcile inverted class gradients.
- Future work should explore class-conditional scaling or Dirichlet calibration (Kull et al., 2019).

4. Reversal of the Rare-Class Hypothesis

Dominant vs Rare Dynamics:

- We hypothesized that miscalibration would track rarity (Twi Normal & Hausa Hate).
- Instead, dominant classes exhibited higher ECE due to inflated confidence from abundant training instances.
- However, rare classes suffer higher operational risk due to high blinding rates (39.6% on Twi Normal).

CONCLUSION & DEFENCE SUMMARY

Key Takeaways and Recommendations for Content Moderation Systems

RULE 1: Never Rely on Aggregate ECE

A single aggregate calibration metric creates a dangerous safety illusion. On Twi, aggregate ECE dropped by -0.013 while minority Normal ECE surged by +42%. Always mandate per-class calibration reporting before deployment.

RULE 2: Audit Post-Hoc Mitigations Per Class

Post-hoc temperature scaling minimizes aggregate NLL, allowing the majority class to hijack the temperature scalar. Recalibration techniques must be validated on every class independently to prevent minority degradation.

RULE 3: Surface Blinding Rates in Triage

System documentation and triage interfaces must visibly report blinding quantities. Human moderation teams must know what fraction of automated decisions conceal confident errors to calibrate review thresholds.

SUMMARY FOR ORAL DEFENCE | READY FOR EXAMINATION QUESTIONS

- Investigating Twi and Hausa under inverted class distributions revealed that calibration does not simply track minority status.
- Majority classes suffer from inflated confidence, and post-hoc temperature scaling can actively worsen minority calibration (+42% on Twi Normal).
- Student: Reynolds Elikem Fenuku (ID: 22032896) | Repository: https://github.com/NEMESIS-7/CMSMM_finalproject
- Thank you. I welcome your questions and oral defence.